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Adaptive digital Electro-Optical Phase-Locked Loop for frequency modulation linearization of FMCW LiDAR system

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Abstract

We demonstrate a closed-loop Electro-Optical Phase-Locked Loop (EO-PLL) designed to increase the linearity of frequency chirps in Frequency-Modulated Continuous-Wave (FMCW) LiDAR systems, resulting in improved metrological accuracy. The system is fully implemented on an easily accessible FPGA-based digital electronic platform (Red Pitaya STEMLab 125-14). A PC-based UI was developed to facilitate remote real-time control over key system parameters. The proposed closed-loop control reduces the Full-Width Half Maximum (FWHM) in frequency domain to 11 kHz, corresponding to a range resolution of 19.9 mm at a beat frequency of 2.45 MHz. Additionally, the system effectively suppresses disturbances in the modulation signal for frequencies up to 150 kHz, with stabilization settling within a single modulation period. This approach enables the use of lasers with nonlinear modulation slopes in FMCW LiDAR applications, even under fluctuating ambient conditions.

1 Introduction

With the advancement of technology for autonomous driving systems, accurate and mass-producible distance and velocity sensors are essential for environmental mapping for driving scenarios [1, 2]. Light detection and ranging (LiDAR) systems based on Frequency-Modulated Continuous-Wave (FMCW) are promising candidates to fulfill this purpose. The main advantages of this technique are reduced sensitivity to interference from surrounding light and lower required laser peak powers due to the use of coherent detection [1, 3]. Furthermore, FMCW LiDAR allows the simultaneous measurement of distance and velocity based on the Doppler effect [4, 5].

Conventional FMCW LiDAR utilizes a triangular-wave frequency-modulated CW laser light, the target detection of which is based on coherent mixing of reflected and

reference optical signals. Accurate target detection relies on the linear frequency chirp of the laser source signal [6]. A low-cost approach to modulating the laser frequency is by direct modulation of the injection current. However, the nonlinear frequency response of the laser cavity leads to a nonlinear frequency sweep using regular triangular current modulation. A high-precision FMCW LiDAR thus requires a linearization mechanism.

One major category involves open-loop pre-distortion, where the laser's modulation waveform is corrected before it is applied. Early work by Karlsson and Olsson demonstrated this by characterizing the laser's frequency response to create a static, pre-compensated waveform [7]. More recent advancements have employed similar strategies [8], some implement more sophisticated methods, such as iterative learning control [9, 10], or interpolation-based algorithms [11], to find a highly optimized pre-distorted waveform. While these methods can achieve impressive linearity, their static nature is a critical drawback; they cannot adapt in real-time to dynamic variations in the laser's performance caused by thermal fluctuations or aging, limiting their robustness in practical applications [12].

A more robust solution is active, closed-loop control, which provides real-time feedback to dynamically linearize the chirp. This is commonly achieved with an Electro-Optical Phase-Locked Loop (EO-PLL), which locks the laser's sweep rate to a stable reference. High-performance

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analog EO-PLLs have been demonstrated, including application-specific analog and digital integrated circuits [13, 14]. However, custom integrated solutions often suffer from inflexibility and high development costs. Consequently, digital EO-PLLs implemented on Field-Programmable Gate Arrays (FPGAs) have become an attractive alternative, offering reconfigurability and powerful digital signal processing capabilities.

A repetitive control approach has been demonstrated [12], with a resolution of 7.28 mm and a settling time within a few modulation periods. The system requires complex custom-designed control system hardware, including a time-to-digital converter on an application-specific PCB. Other approaches utilize closed-loop control to counter slow environmental changes, which is based on static predistortion for systems with low beat signal frequencies (below 100 kHz) [15]. While hybrid systems combine iterative pre-distortion with active feedback to linearize both up and down frequency sweeps [16], their focus remains on compensating for slower drift rather than high-frequency dynamic noise. Our FPGA-based closed-loop EO-PLL overcomes the limitations of traditional analog EO-PLLs and these pre-distortion-based methods in terms of system complexity, slow response, and the need for temperature-sensitive predistortion.

In this work, we present a large-bandwidth closed-loop feedback system for the linearization of the laser frequency modulation (FM). For high beat frequencies on the order

of several MHz, we achieved a reduction of the range resolution to 19.9 mm, a reduction of 17.6 dB compared to the uncontrolled system. We combine this with an easily accessible, low-cost, and flexible hardware platform by implementing the feedback system on a Red Pitaya STEM-lab 125-14 using its FPGA for our fully digital EO-PLL, suitable for rapid prototyping. Apart from the optical components and photodiodes, no further hardware is required. Compared to existing FPGA-based predistortion techniques for FMCW chirp linearization, our system is highly configurable with ultra-low latency. Suppression of nonlinear chirp distortion up to a 150 kHz disturbance bandwidth is possible.

2 FMCW theory

A schematic representation of an FMCW LiDAR system is shown in Fig. 1. f_{u1} and f_{d1} represent the local oscillator signal frequency on the rising and falling edges. f_{u2} and f_{d2} represent the received (i.e., reflected) signal frequency on the modulation up-ramp and down-ramp with the Doppler effect (moving target).

A triangular-wave frequency-modulated laser light is split into two parts, which are used as reference and probe light. The reflected light from the target is mixed coherently with the reference light. When the laser frequency is modulated linearly, the time of flight can be calculated from the

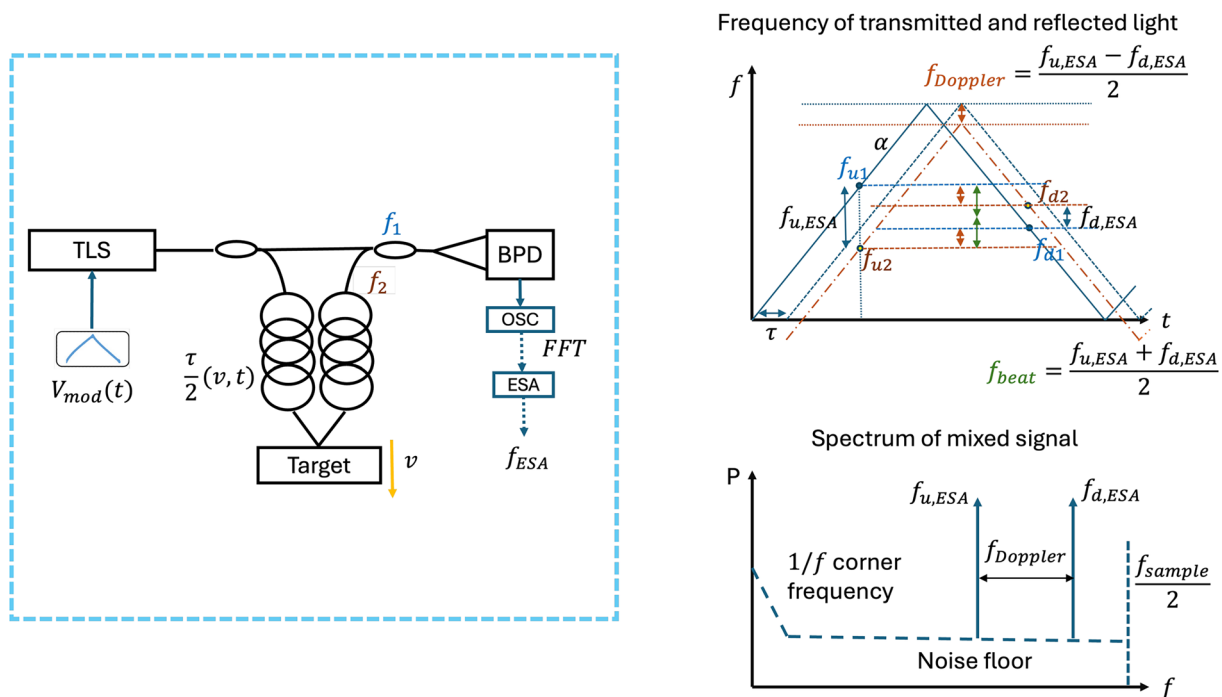


Fig. 1 Schematic representation of an FMCW LiDAR system (left side) including a Tunable Laser System (TLS), Balanced Photodetector (BPD), oscilloscope (OSC), and an Electrical Spectral Analyzer

(ESA), along with the time-dependent frequency of transmitted and reflected light (upper right), and the spectrum of the mixed signal (lower right), figure adapted from [17]

beat frequency between the local oscillator and the reflected light:

$$\tau = \frac{f_{beat}}{\alpha} \quad (1)$$

where α is the chirp rate of the laser source. With this, we can calculate the distance from the target as:

$$d = \frac{c\tau}{2} \quad (2)$$

where c is the speed of light.

Due to the Doppler effect, the triangular frequency curve is shifted up or down with respect to the local oscillator. If the target is non-stationary, the Doppler effect will cause a detected frequency shift of the reflected signal, which leads to two beat notes from the up and down ramps. The target velocity can be calculated as

$$v = \frac{\lambda f_{Doppler}}{2} \quad (3)$$

where $f_{Doppler}$ can be inferred from the frequency shift on the up and down ramps, $f_{u,ESA}$ and $f_{d,ESA}$ respectively:

$$f_{Doppler} = \frac{f_{u,ESA} - f_{d,ESA}}{2} \quad (4)$$

Laser FM nonlinearity can be modeled as $f(t) = f_0 + \alpha t + f_{nl}(t)$, where f_0 is the optical carrier frequency, α is the chirp rate of the laser source, and f_{nl} stands

for the frequency deviation from a linear ramp. The laser FM nonlinearity can be modeled as a periodic modulation signal applied to the FM laser source. According to the Carson bandwidth rule, the bandwidth of the coherent beat note can be calculated as [18, 19]

$$\delta v = 4(1 + 2\pi\tau f_{nl,rms})/T_{mod} \quad (5)$$

where T_{mod} is the modulation period of the laser source and δv is the bandwidth of the beat signal. The modulation index in the expression is simplified by the root mean square of the nonlinearity component of the laser FM. Since the beat signal bandwidth is directly related to the axial resolution of FMCW LiDAR, it is crucial to have a control system that actively corrects the nonlinearity of the laser FM.

3 FPGA-based OPLL

An overview of the control system is shown in Fig. 2 along with the experimental realization in Fig. 3. The initial modulation waveform consists of a triangular wave with an amplitude of 0.1 V and a frequency of 5 kHz. This waveform is fed to a Koheron CTL200 laser control board with a modulation gain of 150 mA/V. The Emcore high-power laser diode mounted on the laser controller has a wavelength of 1550 nm and a linewidth of 10 kHz with a maximum output power of 63.1 mW. The modulation signal has a chirp bandwidth of 5 GHz. The laser light is split and guided into two separate Mach-Zehnder Interferometers (MZIs); one serves to perform the measurement, and one is used as a reference for the feedback. The optical beat signals are converted to

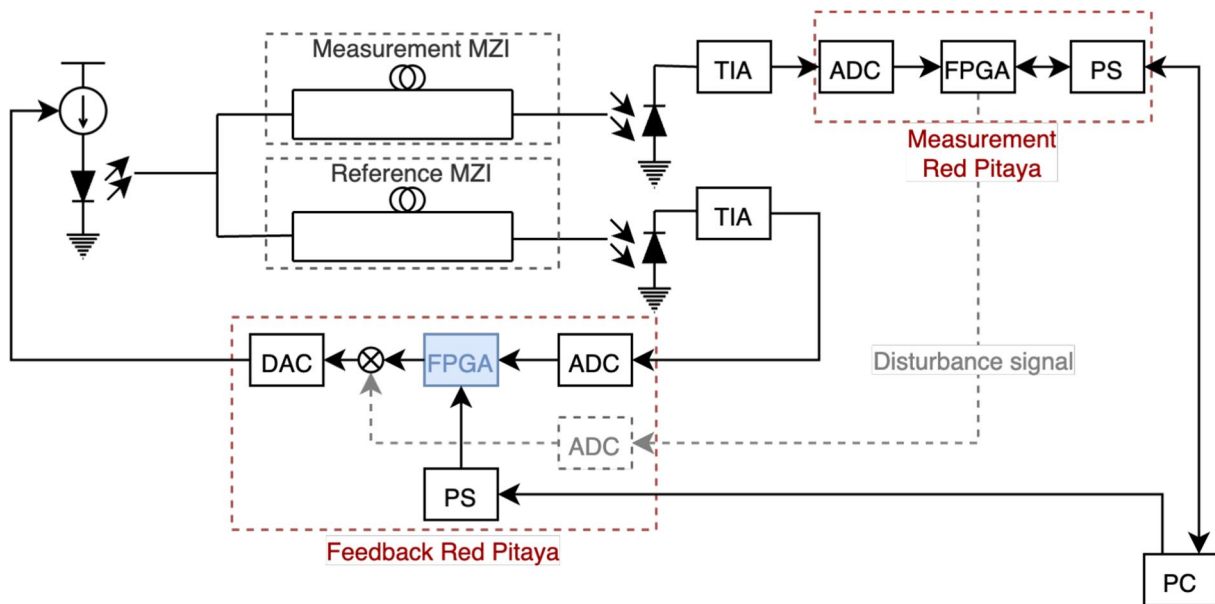


Fig. 2 Schematic representation of the OPLL control loop

result, one period around the turning point is ignored by the Schmitt triggers. The final error signal is generated by subtracting the measured number of cycles from the expected number of cycles per beat signal period. This serves as the input for a PID controller.

As a last step, the modulation waveform needs to be generated. To achieve this, a memory block in the FPGA fabric is configured to store a default triangular waveform. The slope of the output waveform is controlled by changing the step size per cycle of the memory pointer. While the default step size is determined by the set modulation frequency, the step size is dynamically adjusted by an amount corresponding to the output of the PID block. This leads to a higher number of samples of the saved waveform being skipped when the output of the PID is larger and, in an extreme case, causes one sample point to be fed to the output for multiple cycles if the resulting step size is close to zero. A high-precision counter keeps track of the position of the memory pointer and is then rounded to the nearest memory position, which allows for higher precision in the step size. The output waveform is converted to an analog signal by the Red Pitaya's 14-bit Digital-to-Analog Converter (DAC) that has a sampling rate of 125 MHz. A consequence of choosing this approach is that while the beat frequency is controlled to stay nearly constant, the modulation frequency can change if the integral of the PID output over one modulation period is not zero. The amplitude of the modulation waveform, on the other hand, is fixed, except for possible skipped sample points at the turning points of the wave. This differentiates our system from previous active control approaches, where only the absolute value [16] or slope of the output is changed [12, 15], resulting in a constant modulation frequency but a changing modulation amplitude.

In summary, the implementation on the FPGA provides ultra-low latency, consisting of less than a 64 ns delay between a threshold crossing and a reaction at the output.

Table 1 FPGA resource usage of slice Look-Up Tables (LUTs), slice registers, Block RAM (BRAM) tiles, and Digital Signal Processing (DSP) blocks per design component and for the complete design (bold)

Resource	Slice LUTs	Slice registers	BRAM tiles	DSPs
Total available	53,200	106,400	140	220
Usage per component				
Red Pitaya default logic	1116 (2.10%)	1773 (1.67%)	0 (0%)	0 (0%)
Schmitt trigger	237 (0.45%)	499 (0.47%)	0 (0%)	0 (0%)
PID	128 (0.24%)	66 (0.06%)	0 (0%)	3 (1.36%)
NCO	294 (0.55%)	269 (0.25%)	8.5 (6.07%)	0 (0%)
Signal acquisition	30 (0.06%)	71 (0.07%)	3.5 (2.50%)	0 (0%)
Total used	1805 (3.39%)	2678 (2.52%)	12 (8.57%)	3 (1.36%)

Resource usage of the FPGA implementation is analyzed in Table 1, the most critical part being the Numerically-Controlled Oscillator (NCO) (used to store and output the modulation waveform) that uses a considerable amount of Block RAM (BRAM) tiles. As the usage of the BRAM resources of the Red Pitaya FPGA is below 10%, this design allows for scaling up to higher frequencies, higher resolution modulation waveforms, and multiple input and outputs.

For flexibility, the system parameters can be controlled from a user interface on an external PC, as depicted in Fig. 5. Specifically, the modulation frequency and the reference beat frequency can be changed over a network connection without the need to change the hardware on the FPGA. This allows fast adaptation to different setups using a different MZI path delay or modulation frequency. Additionally, the PID parameters can be set using this user interface, facilitating the tuning of the controller. An important feature of the interface is also the adaptability of the default modulation waveform. The memory block within the FPGA can be updated during runtime, making it possible to combine active control with a passive linearization approach using a static predistorted waveform as a starting point for the closed-loop control.

4 Measurement results

To evaluate the performance of the feedback control, we conducted a series of measurements, all of them with a base modulation frequency of 5 kHz and a triangular wave with an amplitude of 0.1 V. This results in a spectral range of 5 GHz for the laser light. The MZIs for the measurements have a path difference of 1.94 m and 4.44 m, resulting in a beat frequency of 1.07 MHz and 2.45 MHz respectively. Ideally, the beat signal used for the feedback is of a higher frequency than the actual signal to be measured. To demonstrate the functioning of the active control also for the stabilization of targets at larger distances, our measurements shown here are performed with the shorter path delay MZI for the feedback path. Increasing the frequency of the reference beat signal with respect to the measured beat signal can increase the bandwidth of the controller, as this effectively increases the feedback rate of the system. In this setup, 9 thresholds are used for the Schmitt triggers, each detecting upward and downward crossings, leading to 18 measurements per period of the beat signal. With a beat signal of 1.07 MHz, this leads to a feedback rate of 19.26 MHz, which already approaches the limit of the system, as the delay between input and output reaction is 64 ns and the time between two feedback values is 52 ns. As the signal processing delay is already dominating the control



Fig. 5 User interface with 4 subsections: **1** Connection control to Red Pitaya, **2** PID parameter control, **3** Signal acquisition and export, **4** NCO and frequency control

system's delay, a further increase in the feedback rate is not expected to improve the performance.

4.1 Phase slope linearity

Ideally, the beat frequency of the measured signal should stay nearly constant over one modulation period T_{mod} . As the frequency of a sinusoidal signal is the derivative of its phase, this is equivalent to the phase of the signal increasing or decreasing linearly. The ideal phase slope can be

calculated as $\gamma_{ideal} = 2\pi f_{beat,ideal}$. Any nonlinearity in the frequency modulation leads to a deviation in the beat frequency $\Delta f_{beat} = f_{beat} - f_{beat,ideal}$ that can be measured as an error in the phase slope $\delta\gamma = 2\pi\Delta f_{beat}$.

The phase slope over one modulation period is depicted in Fig. 6. The turning point of the triangular wave at $0.5T_{mod}$ is clearly visible. Without active control, the phase slope deviates more than 2% from its ideal value for 79.2% of T_{mod} . This can be decreased to 30.3% of T_{mod} with

Fig. 6 Phase slope error $\delta\gamma$ relative to γ_{ideal} over one modulation period, without control (blue) and with control (black); the shaded region shows the standard deviation, $0 \sim 0.5T_{mod}$: down-ramp, $0.5T_{mod} \sim T_{mod}$: up-ramp

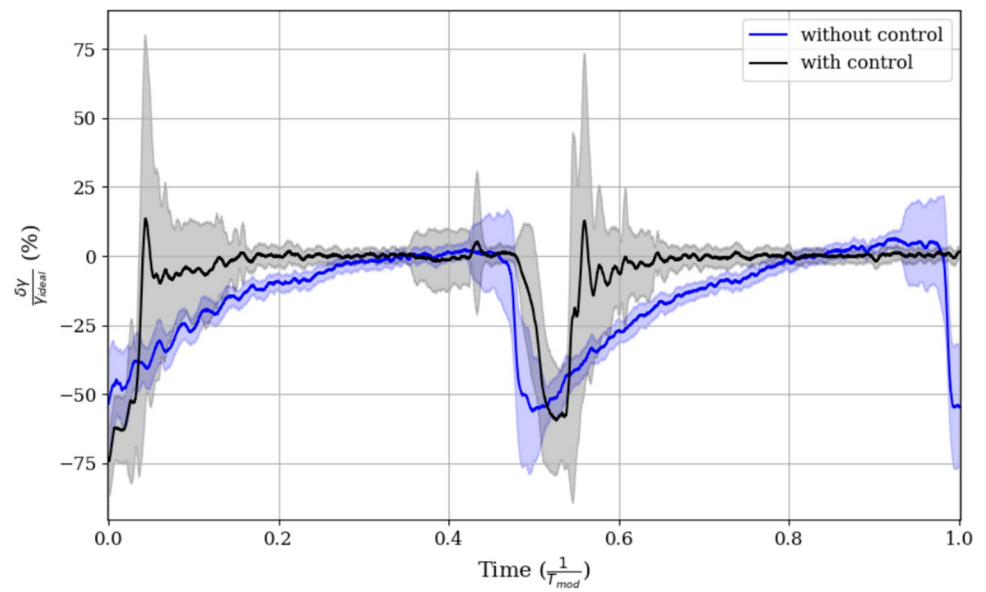
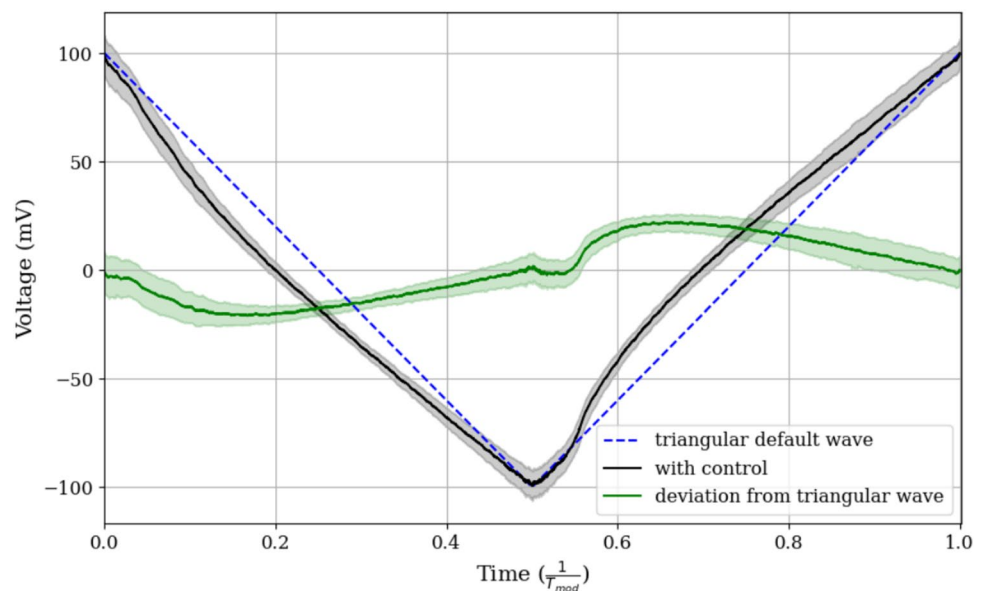


Fig. 7 Modulation waveform over one modulation period (black), triangular wave (blue), and the difference between the two (green); the shaded region shows the standard deviation



feedback control, corresponding to a settling time of the beat frequency of $0.14T_{mod} = 28\mu s$.

Figure 7 shows the modulation waveform with control compared to the default triangular wave. From this, it is clear why there is a transition at the turning points of the triangular wave, where the phase is highly nonlinear. First, the control system misses one period of measurements at the turning point, which corresponds to a relatively large time period because of the lower frequency during this phase. Secondly, the integrator term in the PID control delays the reaction of the system, as shortly before the turning point, the step size is slightly lowered by the feedback, while it has to be sharply increased after the turning point.

As mentioned in Sect 3, this control implementation allows dynamic deviations from the original modulation

period. To quantify this effect, we measured the actual T_{mod} , which yields a standard deviation of $4.86\mu s$ or 2.4% of the expected value.

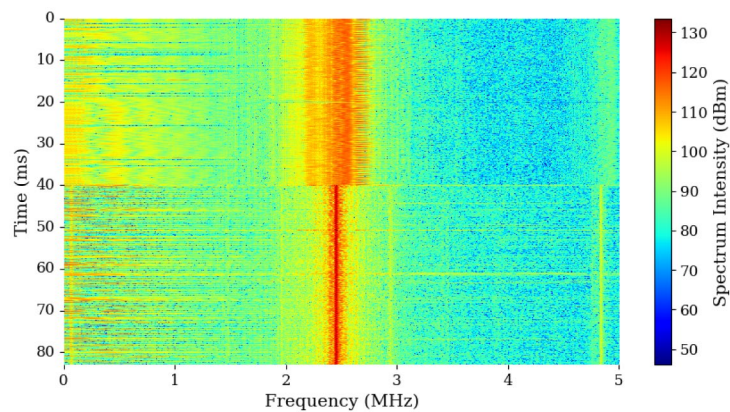
4.2 Distance measurement accuracy

In order to determine the effectiveness of the control system, the influence on the accuracy of the FMCW LiDAR system is crucial. The deciding factor for this is the width of the frequency peak of the beat signal, as the frequency peak is directly proportional to the measured distance.

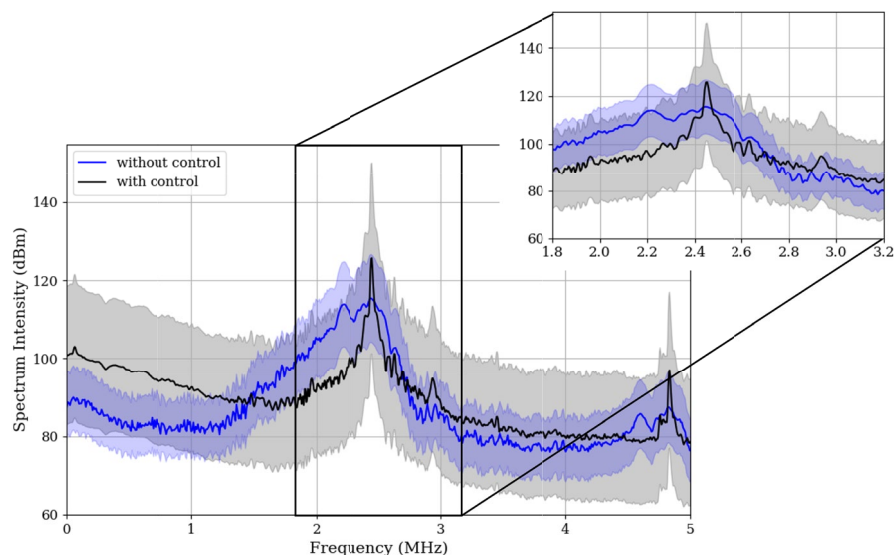
For this, the Fourier Transform (FT) of the beat signal over one modulation period is evaluated. To reduce spectral leakage, a Hann window function was applied before the transform. Figure 8 shows the resulting spectrum of

the measured signal over multiple modulation periods. It is clearly visible that the height of the frequency peak is increased while its width is reduced by the activation of the feedback. Additionally, the narrowing of the peak happens within one modulation period after turning on the control. When activating the EO-PLL, side lobes appear next to the main peak at a distance of 500 kHz. This is caused by the large integrator term of the PID block that causes the main signal to be modulated. Since the side lobes are 30 dBm lower than the main frequency component, this does not significantly affect the measurement accuracy of the FMCW LiDAR. Potentially, these side lobes could be reduced further by using a starting modulation waveform with a more linear frequency chirp instead of the triangular one, as this would cause smaller errors and would thus allow less aggressive PID parameters for similar results.

Fig. 8 Spectrum of the measured beat signal



(a) FT over one modulation period as a function of time, activation of the control at $t = 40$ ms



(b) FT over one modulation period averaged over multiple measurements, without control (blue), with control (black); the shaded region shows the standard deviation

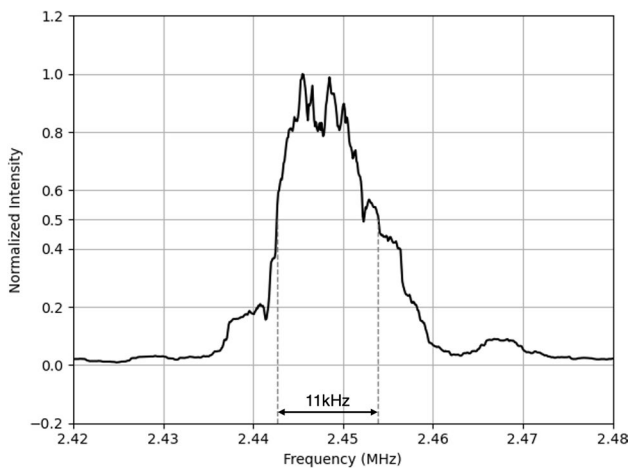


Fig. 9 Spectrum and FWHM around the peak frequency of the FFT calculated over a period of 10 ms or $50T_{mod}$

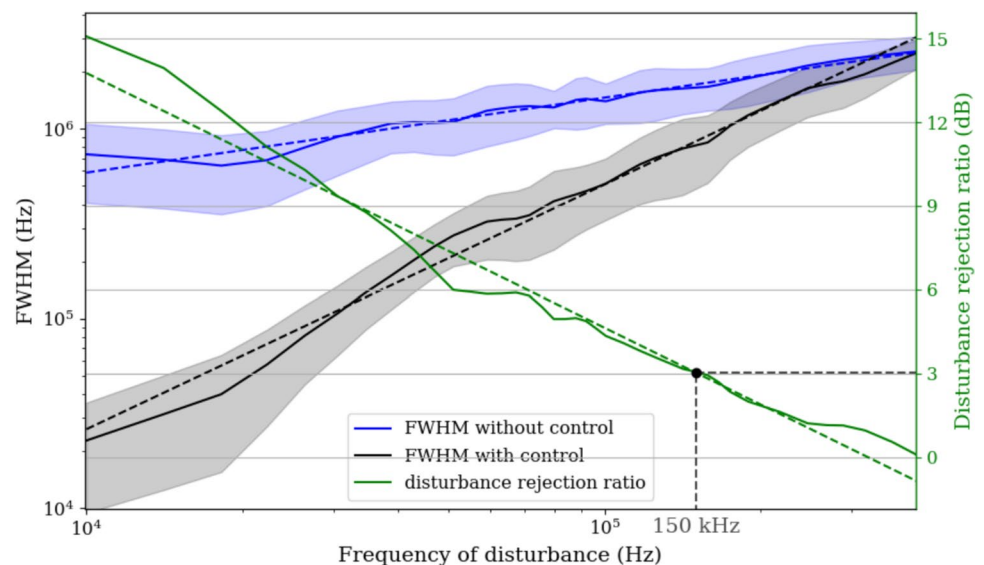
To convert the FWHM in the frequency domain to range resolution at a distance of 4.44 m, the linear relationship between beat frequency and path difference can be used.

$$FWHM_d = FWHM_f \frac{l_{fiber}}{f_{beat, fiber}} = 11 \text{ kHz} \frac{4.44 \text{ m}}{2.45 \text{ MHz}} = 19.9 \text{ mm} \quad (6)$$

4.3 Disturbance rejection and control bandwidth

In the previous sections, the effectiveness of the control system in suppressing the nonlinearity in the transfer function between the voltage input to the laser controller and the frequency output of the laser was evaluated. Not only does this transfer behavior fluctuate due to external influences such as temperature, but using a different laser driver also significantly changes the transfer characteristics. Ideally, the linearization control should be able to suppress external disturbances and support different platforms. To evaluate

Fig. 10 Frequency characteristic of disturbance rejection; an external disturbance at different frequencies is introduced to the modulation waveform and the FWHM without control (blue) and with control (black) is measured; the ratio between the two (green) is a measure for the system's ability to reject the disturbance



the limitations and bandwidth of the active linearization, an external disturbance signal is added to the modulation waveform as described in Sect 3. We vary the frequency of the sinusoidal disturbance and observe the control system's ability to suppress the disturbance by comparing the FWHM with and without control. A measure for the disturbance rejection is the ratio $FWHM_{without_control}/FWHM_{with_control}$. The frequency characteristic of this disturbance rejection ratio is shown in Fig. 10. From this, it can be concluded that the bandwidth of the control system is 150 kHz, above which the FWHM reduction due to the control system becomes lower than 3 dB. At 500 kHz, the FWHM with and without control is equal.

4.4 Performance evaluation

A performance comparison with other state-of-the-art systems is presented in Table 2. While other works have achieved finer range resolutions, our approach provides a competitive resolution of 19.9 mm while offering significant advantages in hardware accessibility and control dynamics. Unlike systems requiring custom PCBs and specialized components, our implementation on a single, low-cost Red Pitaya platform drastically reduces hardware complexity, making it ideal for rapid prototyping. Furthermore, our system is unique in this comparison for demonstrating a large control bandwidth of 150 kHz, which enables fast settling within a single modulation period.

5 Conclusion

In this work, we have presented a closed-loop EO-PLL to increase the linearity of the frequency chirp and the measurement accuracy of the FMCW LiDAR. It is implemented

Table 2 Comparison of FMCW LiDAR systems

	This work	Hauser et al. [12]	Jiang et al. [16]	Müller et al. [15]
Hardware platform	Red Pitaya	FPGA , custom PCB , TDC	FPGA , DAC	FPGA
Linearization technique	Closed loop	Closed loop	Open loop	Closed loop
Laser wave-length (nm)	1550	780	1550	1308
Range resolution (mm)	19.9 (4.4 m distance)	7.28 (3 m distance)	16.8 (0.38 m distance)	2 (0.56 m distance)
Control bandwidth (kHz)	150	N/A	N/A	N/A

fully digitally on a Red Pitaya STEMLab 125-14, making the system affordable, flexible, and suitable for rapid prototyping.

Using closed-loop control, the FWHM in the frequency domain decreased to 11 kHz, which corresponds to a range resolution of 19.9 mm for a beat frequency of 2.45 MHz. The system is also suited to suppress environmental changes in chirp behavior for frequencies of up to 150 kHz, while maintaining its settling time within one modulation period. This allows the use of commercial lasers with nonlinear modulation slopes for FMCW LiDAR in unstable ambient conditions.

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Author contributions Maria Schnuck implemented the FPGA hardware system design in the Redpitaya 125-14 and tested the system performance. She also proposed a new approach for characterizing the control system with artificial noise and Chapters 2, 3, 4, and 5 are written by her. She is the main contributor to the manuscript. Yu Tian proposed the initial system design and helped build the setup for testing the system. He served as the daily supervisor of the project in the Ommatidia Lidar and conceptualized the initial three-layer laser control system. He helped with editing the manuscript. Francisco Diaz Otero is the supervisor of Yu Tian for his Ph.D. project.

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Data availability Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Declarations

Conflict of interest The authors have no Conflict of interest to declare that are relevant to the content of this article.

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